

DBMS COURSE PROJECT • AIML WHALES

Library Circulation & Digital Resource Database

Design and implementation of a 3NF-normalized relational database management system for physical library circulation, digital resources, and automated fines.

LEARNER

Rohan Shikhar

ROLL NUMBER

25WU0101112

INSTITUTION

Woxsen University

Evolution of University Library Operations

Traditional Library Challenges

For decades, academic libraries relied entirely on physical card catalogs and manual ledger books. Counter staff recorded issues and returns by hand, causing friction, book loss, and billing inaccuracies.

- High frequency of human transcription errors.
- Lack of visibility into real-time shelf copy availability.
- Inability to trace multi-author scholarly works efficiently.

The Modern Hybrid Paradigm

Contemporary universities like Woxsen provide both physical textbooks and digital e-journals/e-books. Managing them in isolated silos creates severe administrative blind spots.

- Need for a unified relational system across physical and digital collections.
- Automated tracking of student and faculty circulation policies.
- Real-time fine computation and reservation queue prioritization.

Problem Statement: Why Paper Systems Fail

1. Handwritten & Degraded Records

Entries get copied wrongly, pages tear, and searching old paper registers requires hours of manual leafing. Discrepancies between physical shelf stock and ledger counts are rampant.

2. Zero Real-Time Shelf Visibility

Counter librarians cannot instantly verify if a book copy is available, issued, or reserved without checking physical stacks, leading to long queues and wasted student time.

3. Missed Due Dates & Slipping Fines

Due dates must be checked manually. Overdue loans routinely go unnoticed, fine calculations are inconsistent, and institutional revenue is severely leaked.

4. Isolated Digital Collections

PDFs, journals, and e-books live in disparate links or file folders without central authentication, usage analytics, or member audit trails.

Project Objectives & Deliverables

Objective 1: Automate Circulation Lifecycle

Record every book checkout and check-in as discrete relational transactions in *ISSUE* and *RETURN*, automatically toggling physical copy status between *Available* and *Issued*.

Objective 2: Unified Hybrid Catalog

Integrate physical catalog metadata (ISBN, category, publisher, author) with digital e-resources (PDFs, journals, access URLs) within a single normalized database.

Objective 3: Algorithmic Queue & Fines

Eliminate manual calculations by computing overdue fines strictly from *DueDate* differences and managing reservation waiting queues in first-come, first-served order.

Objective 4: Live Relational Analytics & Web UI

Build a responsive web application connected live to MySQL that enables real-time record viewing, insertion, deletion, and immediate query reporting.

User Roles & Access Permissions

Administrator

Full administrative governance over system data, structure, and reporting.

- ▶ Manage Master Catalog & Publishers
- ▶ Configure Loan Rules & System Parameters
- ▶ Audit Full Digital Access Logs
- ▶ Export Institutional Reports

Librarian (Staff)

Front-counter operational handling of daily circulation activities.

- ▶ Issue and Return Physical Copies
- ▶ Register New Students & Members
- ▶ Collect and Clear Overdue Fines
- ▶ Process Waiting Reservations

Student / Member

End-user patrons seeking library materials and borrowing privileges.

- ▶ Search Available Titles in Catalog
- ▶ Place Reservations on Busy Books
- ▶ Access Online Digital Resources
- ▶ View Personal Borrowing History

Project Scope & Expected Impact

⚡ Operational Efficiencies

- ▶ **Checkout speed:** Book issue and return completed in < 5 seconds via accession numbers.
- ▶ **Zero manual fine computation:** System automatically calculates late fees based on calendar difference.
- ▶ **Strict integrity:** Foreign keys and unique constraints prevent duplicate accession tags and orphan loans.

🔧 Technical Implementation Stack

- ▶ **Database:** MySQL Server 9.7 (InnoDB, Foreign Key enforcement, 13 Normalized Tables in 3NF).
- ▶ **Backend Engine:** Python 3.14 with Flask REST API and PyMySQL connection pooling.
- ▶ **Frontend Interface:** Vanilla CSS & Responsive Modern UI with real-time MySQL synchronization.